

The Five Words — how the machine thinks 五語の巻

CSPS AI Training Workshop · one page, no equations · everything else this week is practice

1 · Five words, one nest

MACHINE LEARNING

Instead of writing rules, show the computer examples and let it find the rules itself.

NEURAL NETWORK

One way to learn from examples: a web of simple calculations, loosely inspired by neurons.

DEEP LEARNING

That web, stacked many layers deep — enough depth to learn language, images, sound.

LARGE LANGUAGE MODEL (LLM)

Deep learning that practised one game trillions of times: **guess the next word**. ChatGPT, Claude, Gemini and Copilot are products wrapped around such models.

The one consequence to remember: a next-word guesser writes fluently whether or not it knows the fact. A plausible lie is not a malfunction — it is the mechanism. That is why every exercise this week ends with *find one thing to fix*.

2 · The engineering ladder — four ways people make these machines useful

Rung	In one sentence	Where you practise it this week
Prompt engineering	Write one instruction well.	The five-part instruction: Role · Task · Context · Constraints · Format (Day 1).
Context engineering	Decide what is on the desk when it answers — the document first, the instruction last, nothing confidential.	"The AI reads only what is on the desk"; the Librarian, who answers only from the shelf.
Loop engineering	Build the circuit: draft → verify → fix → save what worked.	The verification moment; your reusable template and prompt library (Day 1 & 2).
Harness engineering	Wrap the model in memory, supervision and recovery so long jobs survive.	Big labs do it in software. You already do it by hand: your prompt library is the memory, <i>find one thing to fix</i> is the supervisor — you are the harness .

Two fresh examples (August 2026), for the curious: NVIDIA reported that an unchanged frontier model scored ~30% alone on a reasoning benchmark and 100% inside its agent harness (public set, vendor-reported). Google Research showed that adapting the *training environment* around a frozen learner — while keeping the test fixed — beat simply adding more practice. The race has moved from bigger models to better scaffolding. This dojo trains the scaffolding.

Words you may also hear: *token* — the word-pieces the model reads and writes · *hallucination* — the plausible lie above · *agent* — a model that chooses tools and acts, not just answers · *RAG / grounding* — answering only from documents you supply, like the Librarian.